

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Built production **GenAI-powered product features** using **Anthropic Claude, OpenAI GPT-4o, and Google Gemini** at an AI startup -- 10+ features shipped serving real users. Full-stack ownership from DB to React UI; event-driven architecture at **2,000 async req/sec**. Benchmarked Claude vs GPT-4o vs Gemini on 2,000-example test set (hallucination rate, latency, cost). Ready to ship AI-powered ticket discovery experiences at StubHub.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Built **GenAI-powered product features** integrating **Anthropic Claude, OpenAI GPT-4o, and Google Gemini** via MCP -- structured outputs, provider-agnostic interfaces, and agentic workflow orchestration with Temporal.io, serving real users in production.
- Shipped 10+ user-facing features end-to-end using TypeScript, Node.js, Express, React, Prisma ORM, PostgreSQL, Redis, Auth0, Docker, Sentry; real-time event pipelines and PR reviews across the engineering team.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built AI-driven claim automation (OCR, NLP, ML) improving accuracy 25%; automated AWS and GCP deployments with Grafana + Sentry observability.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka + Spark + Hadoop pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

LLM Eval Harness 2025

Vendor-neutral benchmark comparing **Claude, GPT-4o, and Gemini** on 2,000-example frozen test set across grounded QA, multi-hop reasoning, and claim verification. Tracked hallucination rate, p50/p95 latency, cost/call, and abstention accuracy -- bootstrap significance testing and McNemar test for pairwise model comparisons.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

AlgoPulse -- Agentic AI Platform 2025

Event-driven LLM orchestration platform for high-traffic AI workloads. Diagnosed API-polling bottleneck; pivoted to Supabase Edge Functions streaming AI reasoning to client in real time. Temporal.io for durable state + Kafka for fault-tolerant coordination -- **2,000 async req/sec** without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

PulseRank -- Real-Time Leaderboard & Activity Feed 2026

Consumer-facing real-time platform: **Next.js + React** frontend, MySQL outbox pattern, MongoDB TTL feed projections, **Redis sorted sets for sub-ms leaderboard reads**, Server-Sent Events for live score updates. OpenTelemetry + Prometheus observability; GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, React, MySQL, MongoDB, Redis, Docker, Prometheus

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Algorithms, Data Science, Operating Systems, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, Big Data Infra, DS & Algorithms, Data Warehousing

SKILLS

Languages: TypeScript, Python, JavaScript, Java, SQL, C

AI/ML: Anthropic Claude API, OpenAI API, Gemini API, LangChain, Hugging Face, PyTorch, RAG, MCP, LLM Evals

Stack: React, Next.js, NestJS, Node.js, Express, Prisma ORM, Temporal.io, Kafka, Tailwind

Infra: AWS, GCP, Docker, PostgreSQL, Redis, MongoDB, Sentry, Prometheus, GitHub Actions

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)